Tangents and normals

1

a i $y' = \frac{1}{x}$ ii $\frac{1}{5}$

b i $y' = 4 + \frac{1}{x}$ ii 9

c i $y' = \frac{2x}{x^2 + 5}$ ii $\frac{3}{7}$

d i $y' = \frac{4x^3}{x^4 + 4}$ ii $\frac{8}{5}$

e i $y' = 4x \ln x + 2x$ ii $6e$

2

a i $-\frac{2}{5}$ ii $y = -\frac{2x}{5} + \ln 2 + \frac{2}{5}$

b i -5 ii $y = -5x + \ln 5 + 25$

c i $-3 \ln 3$ ii $y = -3x \ln 3 + 9 \ln 3 + 1$

3

a $x = 1$ **b** $y = x - 1$ **c** $y = 1 - x$

4

a $x < 0$ **b** $\frac{dy}{dx} = \frac{1}{x}$ **c** $-\frac{1}{5}$

5

a $y' = \frac{4x}{x^2 + e}$ **b** 0 **c** $y = 2$

6

a $y' = \frac{1}{x \ln x}$ **b** 0 **c** $y = \frac{1}{e}x - 1$

7

a $y' = \frac{9x^2}{x^3 + 2}$ **b** 3

c $y = -\frac{x}{3} + 3 \ln 3 + \frac{1}{3}$

8

a $y' = \frac{-4}{x} + 2$ **b** $y = -0.97$ **c** $x = 2$

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9 $a = -3$

Graphs of logarithmic functions

10

a

i $x < 0$ ii $\frac{1}{x}$ iii $x < 0$

b

i $x > 7$ ii $\frac{1}{x-7} + \frac{1}{x}$ iii $x > 7$

11

a Increasing **b** No

c It decreases at a decreasing rate. **d** It increases towards ∞.

12

a $y = -3\ln(x-4)$ **b** $\frac{dy}{dx} = -\frac{3}{x-4}$

c Undefined when $x \leq 4$

d Strictly decreasing as the derivative is always positive if $x > 4$

13

a $f'(x) = \frac{4}{x}$ **b** 2 **c** $f''(x) = -\frac{4}{x^2}$ **d** -1

e Increasing **f** Concave down

14

a $x > 0$ **b** $\left(\frac{1}{\sqrt{e}}, -\frac{1}{2e}\right)$ **c** Minimum

15

a $x = e$ **b** Maximum

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16

a $x > 0$

b $y' = 1 + \ln x$

c $\left(e^{-1}, -e^{-1}\right)$

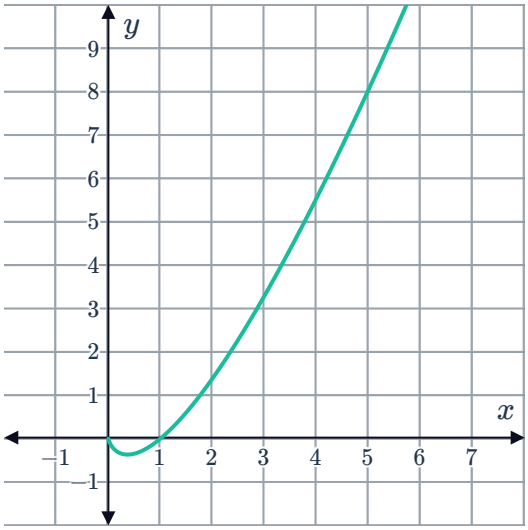
d $y'' = e$

e Minimum

f $y \geq -\frac{1}{e}$

g As $x \to \infty$, $y \to \infty$.

h



17

a $y' = 1 - \frac{1}{x}$

b $(1, 1)$

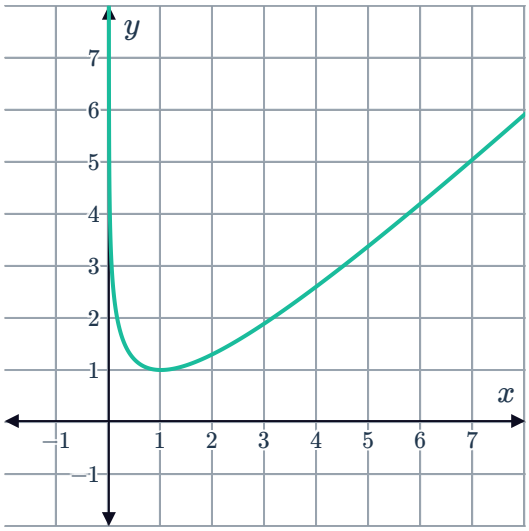
c $y'' = \frac{1}{x^2}$

d Minimum

e As $x \to \infty$, $y \to \infty$.

f As $x \to 0$, $y \to \infty$.

g



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18

a $x = e$

b $(e, 0)$

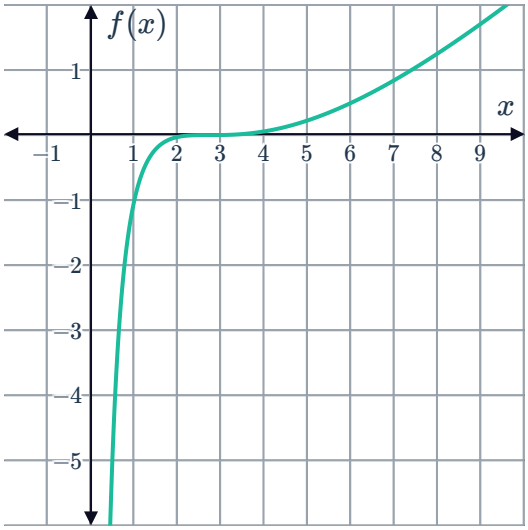
c $f''(x) = \dfrac{3(\ln x - 1)(3 - \ln x)}{x^2}$

d

x	1	e	e^3	e^4
$f''(x)$	-9	0	0	$-9e^{-8}$

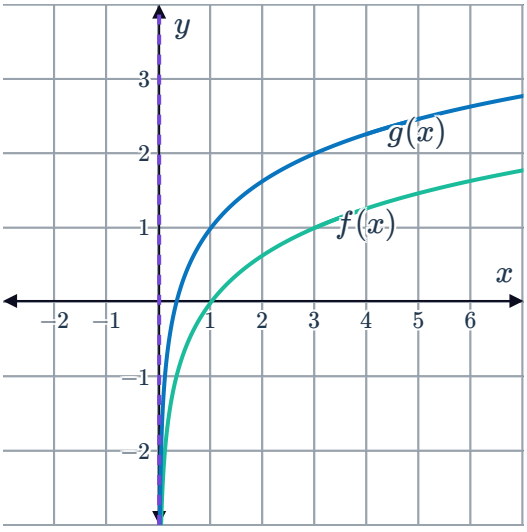
e $(e, 0), (e^3, 8)$

f



19

a



b $f'(x) = \dfrac{1}{x}$

c $g'(x) = \dfrac{1}{x}$

d The tangents to the curve are parallel and increasing at the same rate for any $x > 0$.

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Applications

20

a $\dfrac{dh}{dt} = \dfrac{600}{t+1}$ **b** $120\ \text{m/min}$ **c** Decreasing rate

21

a $m = 3400$. It is the number of passengers serviced in the initial month when $(t = 0)$.

b $k = 150$ **c** $t = 9$ **d** $P'(0) = 150$ **e** $t = 1$ month

22

a $t \geq 0$

b $\dfrac{136}{101}$

c $t = 10$

d **i** $A'(10) = \dfrac{8}{5}$ **ii** $B'(10) = \dfrac{6}{5}$

e Species A

23

a $A = 10.1$ **b** $k = -4.5$ **c** $P'(4) = -0.9$ **d** $t = 9$ years

e $P'(9) = -0.45$

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